

SPLANA

STUDENT TEAM BRIEFING PACK

NDIS Billing Reconciliation Prototype Build

AIH BSIY3008 Business Information Systems | May–June 2026

Prepared for: BSIY3008 Student Team, Australian Institute of Higher Education

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Project Period: 4 May – 26 June 2026 (8 weeks)

Classification: Confidential

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1. Project Context & Your Mission

Welcome to Splana. You are joining us at a pivotal moment: we have validated a significant market opportunity and we are about to begin production development of our software platform. Your prototype will serve as the reference implementation that informs our production build — not a throwaway academic exercise.

1.1 The Problem You Are Solving

Australia's \$45 billion National Disability Insurance Scheme (NDIS) funds disability services for 739,000 participants through 21,000+ registered providers. These providers face a painful operational reality:

- They run rostering and care delivery in one system (typically ShiftCare)
- They run accounting and invoicing in another system (typically Xero)
- They claim payment through the NDIS portal, which has its own rules and rate limits
- These three systems do not talk to each other — data flows through CSV exports and manual spreadsheets
- Admin teams spend 15–30 hours per week manually checking whether shifts match invoices, invoices match NDIS Price Guide rates, and claims are getting paid
- Billing errors, unbilled services, and compliance violations cost mid-sized providers tens of thousands of dollars per year

Splana's platform exists to solve this. Your prototype demonstrates the core reconciliation workflow.

1.2 What We Need From You

Your work has three concrete outcomes:

1. A functional prototype (Objective 1) that demonstrates: CSV upload of rostering data, CSV upload of accounting data, automated matching between the two, and a dashboard that shows discrepancies
2. Professional systems documentation (Objective 2): a System Requirements Document (SRD) and Systems Design Document (SDD) that Splana's future offshore development team can use to build the production version
3. A development roadmap (Objective 3) showing what comes after the prototype — which features to build next, which risks to manage, and how to extend the compliance rules engine

The prototype does not need to be production-grade. It does need to demonstrate the core workflow clearly and produce documentation that is reusable by our production team.

2. User Personas & Key Stakeholders

The prototype has four user types, though only the first two need functional features at this stage. Understanding who will use the system and why is essential to scoping your requirements properly.

2.1 Primary Users (Must Have in Prototype)

Persona	Primary Goal	Typical Context
Provider Finance Manager	Reconcile this week's shifts against invoices; identify underclaimed amounts; prepare NDIS claims accurately	Mid-market NDIS provider (\$5M–\$30M revenue). Uses Xero daily, ShiftCare weekly, exports CSVs to Excel for manual checking
Provider Admin / Billing Officer	Upload CSV exports; identify discrepancies; escalate exceptions to finance manager	Operational staff, moderate technical skill, spends several hours a week on reconciliation. Needs simple workflow

2.2 Secondary Users (Out of Scope for Prototype, but Document in SRD)

Persona	Future Need	Why Documented
Provider CEO / Director	Executive dashboard showing financial health, compliance posture, and recovery opportunities	Board-level reporting needs will drive Phase 2 features
Splana Compliance Consultant	Cross-client view enabling Splana staff to support multiple provider organisations	Important for our consulting-plus-software business model

2.3 User Research Inputs

Splana has completed 13 structured CEO interviews with NDIS providers. Key user insights from this research that should shape your design:

- Users are cautious about technology — technology readiness averages 2.8–3.0 out of 5 across interviews
- Users want "efficiency" and "compliance" framing, not "transformation" or "AI" framing
- Users expect 6-month ROI: if the prototype cannot demonstrate value quickly, it will not be adopted
- Users prefer pilot-first adoption: the prototype should be demonstrable to 5–10 pilot providers
- Users have limited IT support: the interface must be usable by non-technical finance staff

3. Core Workflow: CSV Upload to Reconciliation Dashboard

This section describes the end-to-end workflow your prototype must demonstrate. Think of it as the user journey that the entire prototype is built to deliver.

3.1 Workflow Steps

Step	Action	System Behaviour
1	User logs in	Authenticate with email/password. Role-based access (Admin, Finance User, Read-only). Simple username/password is fine for prototype — no OAuth/SSO needed.
2	User uploads ShiftCare CSV	Parse CSV. Validate column headers. Store shift records with fields: participant ID, service type, date, start time, end time, duration, staff member, location, claim status. Handle common errors (missing fields, date format variations) gracefully.
3	User uploads Xero invoice CSV	Parse CSV. Validate column headers. Store invoice line items with fields: invoice number, line item description, amount, date, participant reference, service code, payment status.
4	System runs reconciliation	Match shift records to invoice line items using participant ID, date, and service type. Flag: (a) shifts with no matching invoice = unbilled; (b) invoices with no matching shift = potential error; (c) amount mismatches between shift duration and invoice amount.
5	System runs compliance check	For each invoice line item, validate amount against NDIS Price Guide rate limits for that service type. Flag line items exceeding limits. Flag line items using incorrect support category codes.
6	User views dashboard	Display: total unmatched shifts (estimated revenue leakage \$ amount), total invoice errors (count + \$ amount), total compliance violations (count + \$ amount), breakdown by participant, breakdown by staff member.
7	User exports report	Generate downloadable CSV or PDF summary showing all flagged items with sufficient detail for admin staff to investigate and correct.

3.2 Matching Logic Recommendations

The matching algorithm is the analytical core of the prototype. Suggested approach:

4. Primary match: exact match on participant ID + service date + service type
5. Secondary match: participant ID + date range (± 1 day) + service type (handles late-billed invoices)
6. Tertiary match: participant ID + service type + month (loose match, flagged for human review)
7. No match: flag as unmatched with recommended action (unbilled shift or potential invoice error)

You do not need to build machine learning or fuzzy matching. Deterministic rule-based matching is sufficient and is what Splana actually wants in production — predictable, auditable, explainable.

4. NDIS Price Guide Compliance Logic

Compliance checking is a key differentiator for Splana. The NDIS Price Guide is a publicly available document that specifies maximum allowable rates for each service type under the scheme. Providers who claim amounts exceeding these rates face payment rejection and potential compliance action.

4.1 Compliance Rules to Implement

Rule Type	Check	Flag Severity
Rate limit compliance	Is the hourly/item rate $\leq$ the NDIS Price Guide maximum for this service code and location?	High (payment rejection risk)
Service category match	Is the service code valid for the participant's plan category?	High (compliance risk)
Duration reasonableness	Is the billed duration within normal operating range (not 24+ hours in a single claim, not zero)?	Medium (data quality)
Time-of-day pricing	Does the rate correctly apply weekday/evening/weekend/public holiday loading?	Medium (underclaiming risk)
Cancellation rule	Are cancellation claims within allowed time limits and participant consent?	Low (edge case)

4.2 Reference Data: NDIS Price Guide

The NDIS Price Guide 2025–26 is publicly available and contains the rate limits you need. Important design note: the Price Guide changes annually and mid-year. Your system must store rates as configurable reference data, not hard-coded values.

Suggested data structure for price rules:

price_rules

- support_item_code (e.g., 01_002_0107_1_1)
- service_description
- registration_group
- unit (hour, each, day)
- rate_standard (weekday)
- rate_evening
- rate_weekend
- rate_public_holiday
- geographic_area (national, remote, very remote)
- effective_from_date
- effective_to_date

This structure allows the Price Guide to be updated without code changes when the NDIS publishes annual price revisions. This is a critical production requirement — not optional.

5. Data Model & Technical Architecture

Splana's target production architecture is Next.js (React) frontend, Node.js backend, PostgreSQL database, hosted on AWS Sydney for Australian data sovereignty compliance. Your prototype does not need to match this stack, but the data model and architectural patterns should mirror the production target so your work is reusable.

5.1 Prototype Stack Flexibility

Acceptable prototype implementations include:

- Python (Flask or FastAPI) + SQLite — fast to build, good for undergrad team
- Python + PostgreSQL — closer to production target
- Node.js + Express + SQLite — if the team has JavaScript skills
- Low-code (Retool, Glide, Bubble) — only if the team explicitly lacks coding capability; not preferred

Recommendation: Python + SQLite with a simple Bootstrap or Tailwind UI is likely the fastest path to a functional prototype for an undergrad team. The database schema and business logic are what matter — they translate directly to our production stack.

5.2 Suggested Data Model

Core entities for the prototype:

shifts

- shift_id, provider_id, participant_id, service_type, service_code, start_datetime, end_datetime, duration_hours, staff_id, location, source_file, uploaded_at

invoices

- invoice_id, provider_id, invoice_number, issue_date, payment_status, total_amount, source_file, uploaded_at

invoice_lines

- line_id, invoice_id, participant_id, service_description, service_code, amount, quantity, line_date

price_rules

- rule_id, support_item_code, service_description, unit, rate_standard, rate_evening, rate_weekend, rate_public_holiday, geographic_area, effective_from_date, effective_to_date

reconciliation_results

- result_id, run_datetime, shift_id, invoice_line_id, match_status (matched/unmatched/partial), discrepancy_type, discrepancy_amount, flag_severity

compliance_flags

- flag_id, invoice_line_id, rule_violated, severity, recommended_action, reviewed, resolved

users

- user_id, email, password_hash, role, provider_id, created_at

5.3 Security & Data Sovereignty Requirements

NDIS participant data is classified as sensitive health information under the Privacy Act 1988. Even for the prototype, document these requirements in your SRD:

- All data must be stored in Australia (AWS Sydney region for production)
- Encryption at rest for database storage
- Encryption in transit (HTTPS) for all data transfers
- Role-based access control with audit logging of who accessed what and when
- Password hashing with modern algorithms (bcrypt, argon2)
- No sharing of participant data outside authorised users
- Compliance with Australian Privacy Principles (APPs)

For the prototype itself, you can simplify (local SQLite is fine, basic authentication is fine) — but the SRD and SDD must document what production-grade security looks like so our offshore development team can implement it correctly.

6. Sample Data I Will Provide

At project kickoff, I will provide de-identified sample data so you do not have to generate synthetic data. Using realistic samples will reveal the edge cases that matter.

6.1 Sample Files You Will Receive

File	Contents	Records
shiftcare_shifts_sample.csv	De-identified shift records from a typical NDIS provider (1 month)	~200–400 shifts
xero_invoices_sample.csv	De-identified invoice exports matched to the shift data period	~50–150 invoices with 400+ line items
ndis_price_guide_2025_26.csv	Structured extract of the public NDIS Price Guide	~1,500+ support item codes
test_scenarios.md	Deliberate edge cases	~20 scenarios

	embedded in sample data: unmatched shifts, rate overruns, duplicate invoices, format irregularities	
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6.2 Known Data Quality Edge Cases to Handle

Real NDIS provider data is messy. The prototype should handle these common issues without crashing:

- Date format variations (DD/MM/YYYY, YYYY-MM-DD, DD-Mon-YYYY mixed within a file)
- CSV encoding issues (UTF-8 BOM, Windows-1252, smart quotes in text fields)
- Missing optional fields (empty location, missing staff ID)
- Duplicate participant names across different IDs (use IDs, not names, as the match key)
- Invoice amounts as strings with currency symbols ("1,234.56" not 1234.56)
- Shifts spanning midnight (start 22:00 day 1, end 06:00 day 2)
- Cancelled shifts that may or may not be billable depending on cancellation rules

Graceful error handling is a core deliverable. Show the user what failed, where, and what to do about it — do not just crash.

7. Dashboard Requirements

The dashboard is the user-facing output of the reconciliation engine. It answers one primary question: "Where is money leaking out of my business?"

7.1 Dashboard Core Views

View	Key Metrics	User Question Answered
Summary / Home	Total revenue leakage (\$), total compliance flags (count), reconciliation coverage (%), last upload date	"How bad is it this week?"
Unmatched Shifts	List of shifts with no matching invoice, participant, date, estimated value	"What did we forget to bill?"
Invoice Errors	List of invoices with no matching shift or with amount mismatches	"What might be incorrectly billed?"
Compliance Flags	List of invoice line items violating NDIS Price Guide rules, with rule cited	"What is at risk of payment rejection?"
Export Reports	CSV/PDF download of any filtered view for investigation	"Can I send this to my admin team to fix?"

7.2 Visual Design Direction

Simple, clean, data-first design. The dashboard will be used by time-pressed finance staff, not executives. Priorities:

- High information density — tables with filters beat charts for operational use
- Dollar amounts prominently displayed (revenue leakage \$ is the headline metric)
- Colour-coded severity (red/amber/green) for flags
- Mobile/tablet responsive — some users will check the dashboard on tablets
- No gratuitous charts — if a table is clearer, use a table

Splana brand colours if you want to align: primary blue #006AFF, dark text #1A1A2E, accent purple #4D4DB4, background off-white #F7F6F2. Use Arial or similar clean sans-serif for body text.

8. SRD and SDD Documentation Requirements

The System Requirements Document (SRD) and Systems Design Document (SDD) are program deliverables due in Week 4. For Splana, these documents are as valuable as the prototype itself — they become the specification our production development team works from. Please invest real effort here.

8.1 SRD Must Cover

- Functional requirements for all four user personas (including the secondary ones not built in the prototype)
- Non-functional requirements specific to NDIS context: Australian data residency, Privacy Act 1988 compliance, audit trail requirements, availability targets for provider operations
- Data input specifications: CSV file formats from ShiftCare and Xero, column definitions, expected record volumes per upload
- Business rules: reconciliation matching logic, NDIS Price Guide compliance rules, exception handling
- Integration requirements (for future phases): direct API integration with ShiftCare and Xero, NDIS portal claim submission, reporting to financial systems
- Constraints: mid-market provider typical infrastructure (limited IT support, Windows-dominant desktop environment, mix of browser versions)

8.2 SDD Must Cover

- Prototype architecture diagram (component view of what you built)
- Production architecture recommendation (what Splana should build next, mirroring the production stack: Next.js + Node.js + PostgreSQL on AWS Sydney)
- Database schema with entity-relationship diagrams
- API design (at least conceptual REST endpoints, even if not fully implemented)
- Security architecture: authentication, authorisation, data protection at rest and in transit
- Deployment considerations: how Splana would host this for real providers, scaling considerations, backup and disaster recovery

8.3 Where to Lean Hard

Because the prototype is low-fidelity by necessity, your documentation is the main way your work compounds into production value. In particular:

- Document the reconciliation matching algorithm with pseudocode and test cases — this is the intellectual property
- Document how the Price Guide compliance rules are structured as configurable data, not hard-coded logic
- Document the edge cases you found in sample data and how the system handles them
- Document assumptions you made and why — this protects production developers from inheriting unstated constraints

9. Deliverable Expectations & Working Rhythm

9.1 Deliverables Checklist

Deliverable	Due Week	Splana's Expectation
Project Plan	Week 2	Scope confirmation, team role allocation, technology stack decision, realistic timeline
System Requirements Doc (SRD)	Week 4	Functional and non-functional requirements covering NDIS-specific context, data residency, Privacy Act 1988
Systems Design Doc (SDD)	Week 4	Prototype architecture plus production architecture recommendation aligned with Splana's target stack
Working Prototype	Week 8	Demonstrable CSV upload, reconciliation, compliance checking, and dashboard with provided sample data
Final Report & Presentation	Week 8	Findings, limitations, and development roadmap for post-project continuation

9.2 Working Rhythm

I will allocate one hour per week to support your team, plus an initial kickoff and two to three mid-project catch-ups. To make this time most useful:

- Send technical questions in writing ahead of catch-ups so I can provide substantive answers
- Share code via a GitHub (or similar) repository so I can see progress between scheduled sessions
- Flag blocked tasks early — I would rather rescope than watch a deliverable slip
- Demo the prototype in small increments rather than one big reveal at Week 8

9.3 What Happens After Your Project

After your project ends, Splana will engage an offshore development team to build the production version of this platform. Your SRD, SDD, and prototype source code will be handed to that team as the starting point. This means:

- Clean, commented code matters — someone else will read it
- Clear documentation of design decisions matters — assumptions buried in code cause production bugs
- Honest documentation of limitations matters — we need to know what you did not solve so we can budget for it
- A working prototype plus excellent documentation is more valuable than a polished prototype with minimal documentation

Your work will be credited in Splana's product documentation and any case study we publish about the university partnership. This is a genuine contribution to a venture building toward commercial launch.

9.4 Resources I Will Provide at Kickoff

- Sample data files (ShiftCare CSV, Xero CSV, NDIS Price Guide structured extract, test scenarios)
- Wireframe concepts showing target dashboard layouts (Figma link)
- This briefing pack plus supplementary context documents
- Access to a GitHub repository for code collaboration
- Email and phone contact for direct questions between scheduled sessions

End of Briefing Pack

For questions, contact Gabriel Gichuhi at grow@splana.com.au